VIT-AP UNIVERSITY

DIGITAL LOGIC DESIGN

(Course Code: ECE 1003)

Module-1 :Lecture-6

Number Systems

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CONTENTS

Lecture-6

- Binary Codes
  - Weighted Codes
  - Non-weighted Codes
- BCD Arithmetic

BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, "**Modern Digital Electronics**", 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Binary Codes

BINARY CODES

- ✓ **Digital circuits and computer system** process data with **0's and 1's** (i.e., in binary formats) because of the bistable (two stable states) nature of the digital systems.
- ✓ However, in actual practice, these circuits are required to handle **numeric, alphabetic or special characters** which will be **converted into binary form**.
- ✓ For a **binary number of n digits**, it may be represented by **n binary circuit elements** whose output must be 0 or 1.
- ✓ Therefore, there are various ways to represent any incoming data or information in a digital system into binary coded form.
- ✓ In general, the number in a digital system or computer are used in coded form.
- ✓ There are two types of binary codes
  1. **Weighted binary codes**
  2. **Non-weighted binary codes**

WEIGHTED BINARY CODES

- ✓ In **weighted binary codes**, **each digit** is associated with **some weights**.
- ✓ Each bit position is assigned a weighing factor and each digit is evaluated by adding the weights of all the ones in the coded combination
- ✓ There are various types of weighted binary codes
  - **Binary Coded Decimal (BCD Code)**
  - **2421 Code**
  - **Excess-3 Code**
  - **84-2-1 Code**

BCD CODES

- ✓ This another way to represent decimal numbers.
- ✓ As this is weighted code, **each successive digits** from right to left represents weights equal to some specified value. Most commonly the **BCD code** is **8421**.
- ✓ To get the equivalent decimal number, we sum the products of these weights with corresponding binary digit.
- ✓ Decimal numbers **0 – 9** can be represented using **4 binary bits** using 8421 code.
- ✓ There are **6 unused combinations** in this coding scheme.

Binary-Coded Decimal (BCD)

Decimal Symbol	BCD Digit
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

Examples

✓ Convert each of these decimal number into BCD code:

1) 35   2) 98   3) 170.2   4) 2469

Solution:

1) 3 → 0011, 5 → 0101

0011 0101

2) 9 → 1001, 8 → 1000

1001 1000

3) 1 → 0001, 7 → 0111, 0 → 0000, . , 2 → 0010

0001 0111 0000 . 0010

4) 2 → 0010, 4 → 0100, 6 → 0110, 9 → 1001

0010 0100 0110 1001

BCD CODES

Examples

✓ Convert the following BCD codes into decimal equivalent:

1) 10000110 2) 001101010001 3) 1001010001110000

Solution:

1) 1000 0110

↓ ↓

8 6

2) 0011 0101 0001

↓ ↓ ↓

3 5 1

3) 1001 0100 0111 0000

↓ ↓ ↓ ↓

9 4 7 0

BCD CODES

Addition Operation of BCD Codes

- ✓ During the addition of two decimal digits using BCD codes, the sum can't be greater than

9 + 9 + 1 (Previous carry) = 19

- ✓ The **range** of the binary sum will be from **0 - 19** i.e., in BCD, it is from **0000 to 00011001** (0000 to 10011 in binary)

Method for BCD addition

- ✓ Add the two BCD numbers, using the rules for binary addition.
- ✓ If the **sum is equal or less than 1001 (9) without any carry**, then the BCD digit is a **valid BCD digit**.
- ✓ If **sum $\geq$ 1010 (10)**, then the result is an invalid BCD digit, then **add 0110 (6) to the binary sum** to get a **valid BCD digit** and can also produce a carry as well.

BCD CODES

Example-1: Perform the following using BCD addition

1) 3 +5 2) 8+9 3) 23 + 42

Solution:

1)

```
   3        0011
 + 5      + 0101
 ----     -------
   8        1000
```

2)

```
   8              1000
 +   9      +     1001
 ----       ---------
  17            1 0001
            +     0110
            ---------
                1 0111
```

$$1 \quad \underbrace{0111}_{7}$$

3)

```
  23       0010  0011
+ 42     + 0100  0010
----     -----------
  65       0110  0101
```

$$\underbrace{0110}_{6} \quad \underbrace{0101}_{5}$$

BCD CODES

Example-2: Perform the following using BCD addition

1) 167 + 482 2) 893 + 487

Solution:

1)

```
                    1
  167         0001 0110 0111
+ 482       + 0100 1000 0010
------      ----------------
  649         0110 1110 1001
          +        0110
          ------------------
              0110 0100 1001
               6    4    9
```

2)

```
                     1    1
  893           1000 1001 0011
+ 487      +    0100 1000 0111
------     -------------------
 1380           1101 0010 1010
         +      0110 0110 0110
         ---------------------
              1 0011 1000 0000
              1  3    8    0
```

BCD CODES

Add the following BCD numbers:

(a) 0011 + 0100 (b) 00100011 + 00010101

(c) 10000110 + 00010011 (d) 010001010000 + 010000010111

The decimal number additions are shown for comparison.

(a)

BCD	Decimal
0011	3
+ 0100	+ 4
0111	7

(b)

BCD		Decimal
0010	0011	23
+ 0001	0101	+ 15
0011	1000	38

(c)

BCD		Decimal
1000	0110	86
+ 0001	0011	+ 13
1001	1001	99

(d)

BCD			Decimal
0100	0101	0000	450
+ 0100	0001	0111	+ 417
1000	0110	0111	867

BCD CODES

Add the following BCD numbers:

(a) 1001 + 0100 (b) 1001 + 1001
(c) 00010110 + 00010101 (d) 01100111 + 01010011

The decimal number additions are shown for comparison.

(a)

```
              1001                                   9
            + 0100                                  +4
              1101    Invalid BCD number (>9)       13
            + 0110    Add 6
     0001     0011    Valid BCD number
      ↓        ↓
      1        3
```

(b)

```
              1001                                   9
            + 1001                                 + 9
        1     0010    Invalid because of carry      18
            + 0110    Add 6
     0001     1000    Valid BCD number
      ↓        ↓
      1        8
```

BCD CODES

(c)

	0001	0110		16
+	0001	0101		+ 15
	0010	1011	Right group is invalid (>9), left group is valid.	31
		+ 0110	Add 6 to invalid code. Add carry, 0001, to next group.	
	0011	0001	Valid BCD number	
	↓	↓		
	3	1		

(d)

	0110	0111		67
	+ 0101	0011		+ 53
	1011	1010	Both groups are invalid (>9)	120
	+ 0110	+ 0110	Add 6 to both groups	
0001	0010	0000	Valid BCD number	
↓	↓	↓		
1	2	0		

WEIGHTED BINARY CODES

- ✓ Other weighted binary codes are:
  - **2421 Code**
  - **Excess-3 Code**
  - **84-2-1 Code**

Four Different Binary Codes for the Decimal Digits

Decimal Digit	BCD 8421	2421	Excess-3	8, 4, −2, −1
0	0000	0000	0011	0000
1	0001	0001	0100	0111
2	0010	0010	0101	0110
3	0011	0011	0110	0101
4	0100	0100	0111	0100
5	0101	1011	1000	1011
6	0110	1100	1001	1010
7	0111	1101	1010	1001
8	1000	1110	1011	1000
9	1001	1111	1100	1111
Unused bit combi- nations	1010	0101	0000	0001
	1011	0110	0001	0010
	1100	0111	0010	0011
	1101	1000	1101	1100
	1110	1001	1110	1101
	1111	1010	1111	1110

WEIGHTED BINARY CODES

- ✓ **Excess-3 code**: these are **self complementing codes**, i.e., 9's complement of the decimal number is obtained by changing 1's to 0's and viceversa.
- ✓ Example:

4 0 2

↓ 9's Complement

$$\begin{array}{r} 9\ 9\ 9 \\ -\ \underline{4\ 0\ 2} \\ 5\ 9\ 7 \end{array}$$

Excess-3 Code

0111 0011 0101

↓ 1's Complement

$$\begin{array}{r} 1000\ \ 1100\ \ 1010 \\ -\ \underline{1000\ \ 1100\ \ 1010} \\ 0101\ \ 1001\ \ 0111 \end{array}$$

5 9 7

NON-WEIGHTED BINARY CODES

- ✓ Weight is not assigned to the bit positions.
- ✓ Different types of non-weighted codes are
  - **Gray Code**
  - **ASCII Code**

GRAY CODES

- ✓ **Gray codes** are generally used during the **conversion of continuous analog data to digital data**.
- ✓ Advantage of gray code over binary number sequence is that only **one bit in the group changes in going from one number to the next**.
- ✓ This code is useful **to avoid the errors or ambiguity** during transformation from one number to the next.

Gray Code

Gray Code	Decimal Equivalent
0000	0
0001	1
0011	2
0010	3
0110	4
0111	5
0101	6
0100	7
1100	8
1101	9
1111	10
1110	11
1010	12
1011	13
1001	14
1000	15

GRAY CODES

Conversion of Binary to Gray code

- ✓ Record the MSB
- ✓ Add this MSB to the next bit. Neglecting the carry if any, record the sum.
- ✓ Continue recording sums until LSB is reached.

Example:

- ✓ Convert $(110111001)_2$ to gray code.

$$1 \xrightarrow{+} 1 \xrightarrow{+} 0 \xrightarrow{+} 1 \xrightarrow{+} 1 \xrightarrow{+} 1 \xrightarrow{+} 0 \xrightarrow{+} 0 \xrightarrow{+} 1$$

$$1 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1$$

GRAY CODES

Conversion of Gray to Binary code

- ✓ Record the MSB
- ✓ Add this binary MSB to the next significant bit of the gray code.
- ✓ Ignoring carries record the result, continue this process until LSB is reached.

Example:

- ✓ Convert gray code (101110101) to binary code.

1	0	1	1	1	0	1	0	1
1	1	0	1	0	0	1	1	0

GRAY CODES

(a) Convert the binary number 11000110 to Gray code.

(b) Convert the Gray code 10101111 to binary.

Solution (a) Binary to Gray code:

$$\begin{array}{ccccccccccccccc}
1 & \xrightarrow{+} & 1 & \xrightarrow{+} & 0 & \xrightarrow{+} & 0 & \xrightarrow{+} & 0 & \xrightarrow{+} & 1 & \xrightarrow{+} & 1 & \xrightarrow{+} & 0 \\
\downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
1 & & 0 & & 1 & & 0 & & 0 & & 1 & & 0 & & 1
\end{array}$$

(b) Gray code to binary:

$$\begin{array}{ccccccccccccccc}
1 & & 0 & & 1 & & 0 & & 1 & & 1 & & 1 & & 1 \\
\downarrow & + & \downarrow & + & \downarrow & + & \downarrow & + & \downarrow & + & \downarrow & + & \downarrow & + & \downarrow \\
1 & & 1 & & 0 & & 0 & & 1 & & 0 & & 1 & & 0
\end{array}$$

ASCII CODES

- ✓ It is used to represent **numbers, special characters, alphabets or symbols**.
- ✓ **ASCII**- **A**merican **S**tandard **C**ode for **I**nformation **I**nterchange
- ✓ ASCII is 7-bit code but most computer manipulate an 8-bit quantity as a single unit.
- ✓ It is used code 128 characters.
- ✓ 94 graphic characters – printed (A-Z, a-z, 0-9, @, !, #, $, *, &, etc.)
- ✓ 34 non-printed characters (e.g., ESC, DEL, SP, etc.)

ASCII CODES

American Standard Code for Information Interchange (ASCII)

	$b_7b_6b_5$							
$b_4b_3b_2b_1$	**000**	**001**	**010**	**011**	**100**	**101**	**110**	**111**
0000	NUL	DLE	SP	0	@	P	`	p
0001	SOH	DC1	!	1	A	Q	a	q
0010	STX	DC2	“	2	B	R	b	r
0011	ETX	DC3	#	3	C	S	c	s
0100	EOT	DC4	$	4	D	T	d	t
0101	ENQ	NAK	%	5	E	U	e	u
0110	ACK	SYN	&	6	F	V	f	v
0111	BEL	ETB	‘	7	G	W	g	w
1000	BS	CAN	(	8	H	X	h	x
1001	HT	EM	)	9	I	Y	i	y
1010	LF	SUB	*	:	J	Z	j	z
1011	VT	ESC	+	;	K	[	k	{
1100	FF	FS	,	<	L	\	l	|
1101	CR	GS	−	=	M	]	m	}
1110	SO	RS	.	>	N	∧	n	~
1111	SI	US	/	?	O	−	o	DEL

ASCII CODES

Control Characters

NUL	Null	DLE	Data-link escape
SOH	Start of heading	DC1	Device control 1
STX	Start of text	DC2	Device control 2
ETX	End of text	DC3	Device control 3
EOT	End of transmission	DC4	Device control 4
ENQ	Enquiry	NAK	Negative acknowledge
ACK	Acknowledge	SYN	Synchronous idle
BEL	Bell	ETB	End-of-transmission block
BS	Backspace	CAN	Cancel
HT	Horizontal tab	EM	End of medium
LF	Line feed	SUB	Substitute
VT	Vertical tab	ESC	Escape
FF	Form feed	FS	File separator
CR	Carriage return	GS	Group separator
SO	Shift out	RS	Record separator
SI	Shift in	US	Unit separator
SP	Space	DEL	Delete

ASSIGNMENT QUESTIONS

Question-1: compute

a) **18 + 82** using BCD addition

a) **184 + 419** using BCD addition

Question-2: Convert gray code 10101010111 to binary

Thank You